Price Suggestions for Airbnb Properties

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Seminar Paper

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Abstract

This seminar paper analyses the determinants of Airbnb property prices in Prague through advanced spatial analysis techniques.

Machine learning methods, including linear regression and random forest, are employed alongside geospatial features and geospatial models to accurately predict Airbnb prices. Spatial lag and autocorrelation models are integrated to account for geographic dependencies impacting property pricing dynamics.

Exploratory data analysis reveals that Airbnb prices in Prague are influenced by property size, neighborhood characteristics, and proximity to Old Town and key amenities. Points of Interest (POIs) analysis highlights significant concentrations around the most expensive part of the city, the Old Town.

Correlation analysis identifies positive relationships between Airbnb prices and factors such as property size (accommodates, bedrooms, beds, bathrooms), and room types. It shows a negative correlation between the Airbnb prices and the distance from the Old Town and positive correlations between the price and the density of nearby Points of Interests. It also reveals that the POIs are highly correlated with each other.

Predictive modeling using Ordinary Least Squares (OLS) regression and Random Forest regression demonstrates the superior performance of Random Forest in capturing non-linear relationships between price and features, evident in lower Root Mean Squared Error (RMSE) and Rooted Median Squared Error (RMeSE). Incorporation of POI features significantly enhances predictive accuracy, underscoring their pivotal role in determining Airbnb prices.

Interestingly, while the addition of spatial dependencies did not improve the Random Forest model compared to only adding the POIs, it enhanced the performance of the Linear Regression model.

Regression analysis identifies key drivers of Airbnb prices in Prague, emphasizing the positive impacts of property size, amenities like bathrooms, and favorable review scores. Property types such as Shared Rooms and Hotels, even if they occur less frequently, also influence pricing. Proximity to the old town proves to be a significant negative factor, while proximity to other POIs such as restaurants and entertainment options influences the price positively.

In conclusion, this study highlights the central role of location, property characteristics and spatial dependencies in shaping Airbnb's price dynamics in Prague. Integrating these factors into predictive models improves our understanding of property valuation in the short-term rental market and provides valuable insights for property owners and hospitality stakeholders.

1 Introduction and Context of This Seminar Paper

1.1 Introduction

The rise of Airbnb, founded in 2008, has revolutionized the short-term rental market, offering travelers an alternative to traditional hotels and providing property owners with a new source of income (Giovanni et al., 2018). Despite initial criticism, Airbnb has grown exponentially, hosting nearly one billion guests worldwide by 2022 (Camatti, Tollo, and Ghilardi, 2024). The platform connects tourists with property owners through a two-sided market model (Oskam, 2019).

Airbnb's success is driven by its ability to offer unique, personalized accommodation options at competitive prices, catering to diverse traveler preferences. The platform's pricing strategy is dynamic, using machine learning algorithms to predict booking likelihood and adjust prices based on market conditions and listing features (Hill, 2015).

This dynamic pricing strategy is especially relevant in popular tourist destinations like Prague. Recently, the average daily rate for Airbnb accommodations in Prague has increased by 25% due to rising tourist numbers. In 2023, Prague welcomed 7.4 million tourists, a 24% increase from the previous year, pushing up prices further (Smith, 2024). The city's historic center and vibrant cultural scene drive high demand for short-term rentals. Proximity to landmarks and transportation hubs significantly impacts pricing.

In this seminar, I analyze the factors influencing Airbnb property prices in Prague with a focus on spatial analysis. By applying machine learning models such as linear regression and random forest, enhanced with geospatial features, I aim to predict Airbnb prices accurately. My approach incorporates spatial lag and spatial autocorrelation to account for geographical dependencies in property pricing. This analysis not only identifies the main drivers of Airbnb pricing but also improves the understanding of spatial dynamics in the short-term rental market.

1.2 Research Goal of This Seminar Paper

This seminar paper explores the predictive capabilities of geospatial machine learning models in estimating Airbnb property prices in Prague. The paper addresses the following research questions:

- **Q1:** How accurately can geospatial machine learning models predict Airbnb property prices in Prague?
- **Q2:** Which features, including geospatial factors, have the most significant impact on predicted Airbnb property prices in Prague?

By answering these questions, this paper aims to provide valuable insights into the effectiveness of geospatial machine learning models in the context of Airbnb price prediction. Through an examination of various features, including geospatial factors, it wants to identify key determinants of Airbnb property prices in Prague, thus contributing to a deeper understanding of the dynamics shaping the city's short-term rental market.

2 Exploratory Data Analysis

2.1 Data Overview

In this seminar paper, I employed a dataset comprising Airbnb properties in Prague obtained from Cox, Morris, and Taylor, 2024. It was collected on March 25, 2024, and contains 8,029 observations with a variety of numerical and categorical attributes related to the properties, including price, location, property type, room type, availability, and other relevant features.

I merged the listings' data from the website with the neighborhood's file to incorporate geospatial information into the analysis.

2.2 Utilized Property Features

The analysis focuses on a selection of key features extracted from the Airbnb dataset, providing insights into the factors that may influence property prices in Prague. These features include both numerical and categorical variables, which are explained in detail in Table 1.

2.3 Data Cleaning

To ensure the quality and integrity of the dataset, several data-cleaning procedures were conducted. These steps include:

1. Price Variable Currency Transformation
2. Handling Missing Values
3. One-Hot Encoding

2.3.1 Price Variable Currency Transformation

The Inside Airbnb Website claims that the prices are in Dollars. However, after examining the data, it was discovered that the prices were in Czech Koruna (CZK). To ensure consistency and ease of interpretation, the price variable was transformed from the original Czech Koruna (CZK) currency to Euro (€). The conversion rate used was 1 CZK = 0.04 €.

2.3.2 Handling Missing Values

Missing values within the dataset were addressed through imputation with suitable values. Moreover, the missing values of the binary variables were addressed by setting the missing variable to zero.

Feature	Description
Numerical Features	
Price	The nightly rental price of the Airbnb properties.
Location	Geographic coordinates (longitude and latitude) indicating the property's position within Prague.
Accommodates	The number of guests the property can accommodate.
Bedrooms	The number of bedrooms in the property.
Beds	The number of beds available in the property.
Bathrooms	The number of bathrooms in the property.
Review Scores Rating	The average rating of the property based on guest reviews.
Number of Reviews	The total number of reviews received by the property.
Categorical Features	
Property Type	Categorization of properties into types such as apartment, house, or condominium.
Room Type	Classification of rooms available for rent, including entire homes/apartments, private rooms, or shared rooms.
Neighborhood	Information about the neighborhood where the property is located.

Table 1: Features of Airbnb Properties in Prague

2.3.3 One-Hot Encoding

Categorical variables like property types and room types were incorporated into the analysis using one-hot encoding, converting them into numerical format.

2.4 Descriptive Analysis of Property Features

2.4.1 Price Distribution

As the table 2 shows, the mean price of Airbnb listings in Prague is 178.26 €, with a standard deviation of 1608.71 €. The minimum price of an Airbnb per night is 9.12 €. The median price is 100.68 €, indicating that half of the listings are priced below this value.

The table 2 also illustrates, that the maximum price of an Airbnb per night is 93827.96€. This is an unusually high price for an Airbnb listing in Prague.

A threshold value corresponding to the 99.5th percentile was therefore set to define outliers.

Any observations exceeding this threshold were identified as outliers and removed from the dataset. This process resulted in the elimination of 41 outlier instances. The threshold

Table 2: Summary Statistics of Airbnb Prices in Prague

Statistic	**Value (€)**
Mean	178.26
Standard Deviation	1608.71
Minimum	9.12
25% Quantile	66.64
Median	100.68
75% Quantile	157.72
95% Quantile	312.92
Maximum	93827.96

value was 1128.60€.

After removing the outliers, the price distribution of Airbnb listings in Prague was studied. The histogram in figure 2 illustrates the distribution of prices, showing a right-skewed pattern, with a high concentration in the lower range.

The spatial representation in Figure 3 illustrates the spatial distribution of prices. It shows that Airbnb accommodation in the old town, close to the main tourist attractions, tends to be priced higher, reflecting the influence of location on price dynamics. Furthermore, accommodations located outside the city center show more affordable price points, indicating the varied pricing structures across different regions within Prague.

2.4.2 Overview of Neighborhoods

The analysis of Airbnb properties across various neighborhoods in Prague gives valuable insights into the distribution of listings and pricing trends.

Neighborhoods with more than 50 listings were compared in figure 4 and figure 5 to identify trends in pricing.

Among these neighborhoods, Praha 1 is the most expensive, considering the median price. It is followed by Praha 8 and Praha 2. In districts further away from the city center, such as Praha 10, median prices are lower, reflecting the influence of location on the price.

Considering the number of listings, Praha 1 has the highest number of Airbnb listings, followed by Praha 2 and Praha 3.

This concentration of listings in the city center aligns with the high demand for accommodation in central locations.

2.4.3 Overview of Property Types and Room Types

The Airbnb properties are classified into distinct types to understand further the dynamics of the short-term rental market. This classification includes room types and property types.

The figures 6 and 7 illustrate the most listed room types and property types, as well as their average prices.

Room types include Entire home/apartment, Private Room, and Shared room. As figure 6 illustrates, Entire home/apt emerges as the most listed room type, followed closely by Private Room. Shared Room has the least listings.

The most expensive room type in figure 7 is the Entire home/apt, reflecting the premium placed on exclusive accommodations. In contrast, "Shared Room" represents the most economical option, catering to budget-conscious travelers.

Property types cover a wide range of options, including Condos, Apartments, House, Rental Units, Hotel, and Hostel. As figure 6 shows, the Rental Unit is by far the most common property type, followed by Condos and Apartments. In contrast, houses and hostels have fewer listings, which indicates fewer offers on the platform.

The most expensive property type is Apartment, followed by Hotel, while the property type Hostel represents more budget-friendly options within the Airbnb market.

As the plot in figure 8 illustrates, rental units are common in the city center as well as its outskirts, reflecting their prevalence as the most common property type. Hostels are concentrated in the city center, catering to budget-conscious travelers seeking affordable accommodation. Condos primarily populate areas outside the city center, offering residential tranquility.

The majority of room types cluster within the city center, aligning with the high demand for accommodation in central locations. While Entire Homes/Apartments dominate the urban landscape, they also extend into suburban areas, catering to diverse guest preferences. Private Rooms and Shared Rooms are mostly listed in the Old Town.

2.4.4 Frequency of Property Features

The frequency distribution of property features in figure 9 provides valuable insights into the characteristics of Airbnb listings, giving more information on most accommodation capacities, amenities, and pricing dynamics.

The y-axis of the histograms illustrates the number of listings falling into each category, while the x-axis delineates the individual categories of the columns representing the listings' attributes.

- **Accommodates:** Most listings can accommodate 1-2 guests, followed by 4-5 guests.
- **Bathrooms:** The majority of listings feature 1 bathroom, with 2 bathrooms being the next most common.
- **Bedrooms:** Most listings have 1-3 bedrooms.
- **Beds:** Most listings offer 1 - 3 beds.
- **Price:** represents the frequency of different price ranges for the listings. Most listings have prices between approximately 100€ and 250€ per night.

2.4.5 Correlation Analysis

The heatmap in figure 10 visualizes the correlation coefficients between the price variable and other features.

The key correlations identified in the heatmap include:

- Strong positive correlations between price and accommodates (0.49), bedrooms(0.41), beds(0.3), and bathrooms(0.16), indicating that larger properties tend to command higher prices.
- Some features like accommodates, bedrooms, beds, and bathrooms are highly correlated with each other, indicating multicollinearity.
- Some property types like Apartment(0.12) and Hotel(0.04) have a positive correlation with price, while others like Hostel (-0.04), Rental unit (-0.04), House(-0.01), Condo(-0.07) have a negative correlation.
- Room types like Entire home/apt (0.17) have a positive correlation with price, while Private Room (-0.15) and Shared Room (-0.12) have a negative correlation.

3 Feature Engineering

3.1 Creation of New Geospatial Features

In advancing the analytical capabilities of machine learning models, the creation of geospatial features turns out to be a crucial step. These features allow for the integration of location-based data into predictive models, enabling a deeper exploration of spatial relationships between Airbnb properties and key points of interest (POIs) in Prague.

3.1.1 Extracting Points of Interest Using the OSMnx Package

To capture the spatial distribution of essential amenities in Prague, I utilized the OSMnx Python package. It facilitated the extraction of points of interest (POIs) across the city, using OpenStreetMap data. Data from Boeing (2024).

These POIs are categorized into four distinct groups:

- **Dining and Entertainment:** Encompassing restaurants, bars, fast-food establishments, and cafés.
- **Shopping:** Encompassing shops, marketplaces, and food courts.
- **Sightseeing:** Encompassing tourist attractions, museums, zoos, theme parks, galleries, monuments, and historical sites.
- **Transportation Hubs:** Encompassing bus stations, taxi stops, car rental services, and car-sharing facilities.

These selections were made based on their potential impact on property desirability and potential correlations with pricing trends.

Table 3: Counts of Points of Interest in Prague

POI Category	**Count**
Shopping	2654
Sightseeing	2459
Dining and Entertainment	2066
Transportation	14

As illustrated in figure 11 and table 3, the distribution of POIs across Prague is as follows:

The highest counts are observed in the Shopping category, followed by Sightseeing Dining, and Entertainment. Transportation hubs exhibit the lowest count of only 14 locations, indicating a limited effect of this category on the model.

When looking at the spatial distribution of figure 12 of these POIs in Prague, a clear concentration of POIs is observed in the city center, particularly in the Old Town area. This aligns with the high density of Airbnb listings in this region, as illustrated in figure 13. This Old Town not only hosts major tourist attractions, but also boasts a vibrant array of dining, entertainment, and shopping options.

The high density of POIs in the City Center area aligns with the areas characterized by the highest Airbnb prices, reflecting the premium placed on proximity to cultural landmarks and urban conveniences.

While a substantial concentration of amenities is observed in the historic Old Town area, Peripheral areas of the city exhibit a sparser distribution of amenities. Here, the availability of dining, shopping, and entertainment options tends to be more limited. Consequently, property prices in these areas typically reflect this reduced accessibility to urban amenities, resulting in comparatively lower Airbnb rates.

In essence, the spatial visualization underscores the connection between amenities, property values, and tourist attractions, highlighting the significance of location as a key determinant of property desirability and pricing dynamics in Prague.

3.1.2 Kernel Density Estimation (KDE) for Points of Interest

One of the key geospatial features created for the analysis is the Kernel Density Estimation (KDE) of points of interest (POIs) in Prague. KDE is a non-parametric method used to estimate the probability density function of a random variable and is particularly useful for estimating and visualizing the spatial distribution of POIs across the city, as mentioned in VanderPlas, 2016, Chapter: Kernel Density Estimation. I used the implementation of KDE in the Python library SciPy Virtanen et al., 2020 to generate density plots for each category of POIs. This implementation is based on the Gaussian kernel function, which assigns weights to each POI based on its proximity to other POIs. A bandwidth parameter was selected to control the smoothness of the KDE plots, with a higher bandwidth resulting in a smoother density distribution. The bandwidth parameter was selected automatically using Scott's method. This density information is later used as a feature in the machine learning models to capture the spatial distribution of amenities and its impact on Airbnb prices. Areas with higher KDE values indicate a higher concentration of POIs, which should be reflected in higher Airbnb prices due to the increased convenience and accessibility of amenities. On the other hand, areas with lower KDE values indicate a lower concentration of POIs, which may be associated with lower Airbnb prices.

The KDE plots in figure 14 show that the Density of Airbnb Listings, Sightseeing Spots, Dining and Entertainment Places, and Shopping places is very high. The Density of Transportation Hubs is relatively low. This might be because of the low number of Transportation Hub POIs.

The KDE plots in figure 15 illustrate the density distribution of amenities across Prague, providing insights into the concentration of key POIs in different regions of the city. It shows that the POIs are mostly concentrated in the city center, particularly in the Old Town area, reflecting the high density of amenities in this region.

3.1.3 Calculating Distances from Airbnb's to Old Town

The distance to Prague's Old Town serves as a crucial metric in this analysis due to its centrality to the city's major sightseeing attractions. By measuring the proximity of Airbnb listings to this historic district, insights into their accessibility to key tourist destinations are gained. To calculate the distance from each Airbnb property to the Old Town, I used the geodesic distance formula, which is implemented in the Python library geopy GeoPy, 2024.

3.1.4 Correlation of POIs and Price

The Correlation Matrix 10 reveals several correlations between points of interest (POIs) and Airbnb listing prices. The key correlations identified include:

- A negative correlation (-0.17) between Airbnb listing prices and their distance from the Old Town. This suggests that properties closer to the city's historic center tend to command higher prices, reflecting the desirability of central locations for tourists.
- A positive correlation between the price and the KDE of Airbnbs (0.18), indicates that areas with denser concentrations of Airbnb listings tend to exhibit higher prices, likely because of the high concentration of Airbnbs in the city center.
- Positive correlations between the price and the KDE of Sightseeing Places (0.14), Dining and Entertainment Places (0.12), Shopping Places (0.11), and Transportation Hubs (0.11). These correlations suggest that areas with denser concentrations of these POIs tend to exhibit higher Airbnb prices, likely due to increased convenience and accessibility for guests.
- Strong positive correlations between the KDE Airbnbs and the KDE of sightseeing spots (0.82) and dining/entertainment venues (0.83). These findings underscore the proximity of Airbnb properties to these attractions, highlighting their role in shaping property demand and pricing dynamics.

- The POIs are correlated to each other, indicating that areas with high concentrations of one type of amenity tend to have high concentrations of others as well.

Overall, the analysis suggests that proximity to popular POIs significantly influences Airbnb listing prices, emphasizing the importance of location-based amenities in property valuation.

4 Model Development

This section details the various machine-learning models employed to predict Airbnb prices in Prague. Each model's theoretical foundation, practical implementation, and mathematical formulations are discussed, highlighting their strengths and the reasoning behind their selection.

4.1 Ordinary Least Squares Regression

Ordinary Least Squares (OLS) Regression is a fundamental technique in statistical modelling used to estimate the relationships between a dependent variable and one or more independent variables. The method minimizes the sum of the squared differences between observed and predicted values, as detailed in Rey, Arribas-Bel, and Wolf, 2023, Chapter: Spatial Regression.

The OLS regression equation is given by:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

where:

- y is the dependent variable (Airbnb price),
- β_0 is the intercept,
- $\beta_1, \beta_2, \ldots, \beta_p$ are the coefficients of the independent variables,
- $x_1, x_2, \ldots, x_p$ are the independent variables (features),
- ϵ is the error term.

The features used for the OLS regression include both property-specific characteristics and geospatial features, as detailed in Table 4.

By conducting separate regressions with property features alone and with both property and POI features, I aim to understand the added predictive power of geospatial variables.

4.2 Spatial Lag Model

The Spatial Lag Model incorporates spatial dependencies directly into the regression framework by including a spatially lagged dependent variable. This model is useful when the value of the dependent variable in one location is influenced by the values in neighbouring

Table 4: Features for OLS Regression

Feature Type	Features
Property Features	Accommodates Bedrooms Beds Bathrooms Room Types Property Types Review Scores Number of Reviews
POI Features	KDE of Airbnb's KDE of Sightseeing KDE of Transportation KDE of Shopping KDE of Dining and Entertainment Distance to Old Town (km)

locations. Further details can be found in Rey, Arribas-Bel, and Wolf, 2023, Chapter: Spatial Regression. So in our case, it is assumed that the price of an Airbnb listing is influenced by the average price of nearby listings. The Spatial Lag Model can in this case be formulated as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \rho \overline{y} + \epsilon$$

where:

- y is the dependent variable (Airbnb price),
- β_0 is the intercept,
- $\beta_1, \beta_2, \ldots, \beta_p$ are the coefficients of the independent variables,
- $x_1, x_2, \ldots, x_p$ are the independent variables (features),
- ρ is the spatial autoregressive coefficient,
- $\overline{y}$ is the average price of the k listings nearest to the location of the Airbnb listing in question,
- ϵ is the error term.

4.3 Spatial Cross-Linear Regression

The Spatial Cross-Linear Regression model extends the Spatial Lag Model by incorporating both spatially lagged dependent variables and spatially lagged independent variables. This model is particularly useful when the value of the dependent variable is influenced by both its own and neighbouring values, as well as the values of neighbouring independent variables, see Rey, Arribas-Bel, and Wolf, 2023, Chapter: Spatial Regression.

So in this case, it is assumed that the price of an Airbnb listing is not only influenced by its own property features and POI features but also by the prices as well as the property features of nearby listings.

The Spatial Cross-Linear Regression can in this case be formulated as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \rho \overline{y} + \lambda_1 \overline{x_1} + \lambda_2 \overline{x_2} + \cdots + \lambda_p \overline{x_q} + \epsilon$$

where:

- y is the dependent variable (Airbnb price),
- β_0 is the intercept,
- $\beta_1, \beta_2, \ldots, \beta_p$ are the coefficients of the independent variables,
- $x_1, x_2, \ldots, x_p$ are the independent variables (the q property features and the $p-q$ POI features),
- ρ is the spatial autoregressive coefficient,
- $\overline{y}$ is the average price of the k listings nearest to the location of the Airbnb listing in question,
- $\lambda_1, \lambda_2, \ldots, \lambda_q$ are the coefficients of the spatially lagged property features,
- $\overline{x_1}, \overline{x_2}, \ldots, \overline{x_q}$ are the average property feature values of the k listings nearest to the location of the Airbnb listing in question.

4.4 Random Forest Regressor

The Random Forest Regressor is an ensemble learning method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees. This model is particularly effective in handling non-linear relationships and interactions between variables, (Breiman, 2001). I included the Random Forest Regressor in our analysis to capture the complex relationships between property features, POI features, and Airbnb prices in Prague and see if it can help us to get a larger accuracy in our predictions when including spatial lags and POI features.

For the Random Forest Regressor, the features as for the linear regression models are used:

- Property Features
- Property Features and POI Features
- Property Features, POI Features, Spatial Lag of Price
- Property Features, POI Features, Spatial Lag of Price, Spatial Lag of Property Features

5 Model Evaluation

5.1 Evaluation Approach

5.1.1 Train-Test Split

To evaluate the performance of the models, I divided the dataset into training and testing sets. This ensures that the models are tested on unseen data, providing a realistic measure of their predictive power. The training set was used to fit the models, while the test set was reserved for evaluating their performance.

5.1.2 Spatial Lag Determination

Determining the optimal value of k for the spatial lag model involved cross-validation. The k-value represents the number of nearest neighbours considered for the spatial lag effect. During cross-validation, multiple values of k were tested, and the one that minimized the Root Mean Squared Error (RMSE) across the validation folds was selected. This approach ensures that the spatial lag effect is optimized, improving the model's accuracy.

5.2 Evaluation Metrics

The performance of the machine learning models was assessed using the following evaluation metrics:

5.2.1 Root Mean Squared Error

The Root Mean Squared Error (RMSE) is a standard metric for evaluating the accuracy of predictive models. It measures the average magnitude of the errors between the predicted and actual values. The formula for RMSE is:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where:

- y_i are the actual values,
- $\hat{y}_i$ are the predicted values,
- n is the number of observations.

RMSE was chosen because it provides a straightforward interpretation of prediction errors in the same units as the dependent variable (Airbnb prices), making it easier to understand the extent of errors in monetary terms, (Hodson, 2022).

5.2.2 Root Median Squared Error

The Root Median Squared Error (RMeSE) is similar to RMSE but uses the median of the squared errors instead of the mean. The formula for RMeSE is:

$$\text{RMeSE} = \sqrt{\text{median}((y_i - \hat{y}_i)^2)}$$

RMeSE is particularly useful in the presence of outliers because the median is less sensitive to extreme values than the mean. By using RMeSE alongside RMSE, the reader gains a more robust understanding of model performance, especially in this dataset where Airbnb prices have a right-skewed distribution with potential outliers, as illustrated in figure 2.

5.3 Results and Analysis

Index	Model	RMSE	RMeSE	Rel. Impr. RMSE (%)	Rel. Impr. RMeSE (%)
Linear Regression (Properties)	Linear Regression	79.970000	33.850000	0.000000	0.000000
Linear Regression (Properties + POIs)	Linear Regression	78.010000	31.230000	2.450000	7.740000
Linear Regression (Properties + POIs Independent)	Linear Regression	78.240000	32.110000	2.160000	5.130000
Random Forest (Properties)	Random Forest	82.250000	31.610000	-2.850000	6.610000
Random Forest (Properties + POIs)	Random Forest	69.050000	23.220000	13.660000	31.410000
Random Forest (Properties + POIs Independent)	Random Forest	68.770000	24.170000	14.010000	28.590000
Linear Regression (Properties + Spatial Lag)	Linear Regression	75.090000	29.940000	6.110000	11.550000
Linear Regression (Properties + POIs + Spatial Lag)	Linear Regression	74.600000	29.120000	6.720000	13.990000
Random Forest (Properties + Spatial Lag)	Random Forest	71.830000	25.210000	10.180000	25.530000
Random Forest (Properties + POIs + Spatial Lag)	Random Forest	69.550000	22.830000	13.040000	32.560000
Linear Regression (Properties + Spatial Correlation)	Linear Regression	79.270000	32.910000	0.880000	2.770000
Linear Regression (Properties + POIs + Spatial Correlation)	Linear Regression	77.730000	31.150000	2.800000	7.970000
Linear Regression (Properties + Spatial Lag + Spatial Correlation)	Linear Regression	73.830000	27.760000	7.680000	18.000000
Linear Regression (Properties + POIs + Spatial Lag + Spatial Correlation)	Linear Regression	73.510000	27.680000	8.080000	18.240000
Random Forest (Properties + Spatial Correlation)	Random Forest	74.000000	27.330000	7.460000	19.270000
Random Forest (Properties + POIs + Spatial Correlation)	Random Forest	69.750000	23.650000	12.790000	30.130000
Random Forest (Properties + Spatial Lag + Spatial Correlation)	Random Forest	71.830000	25.210000	10.180000	25.530000
Random Forest (Properties + POIs + Spatial Lag + Spatial Correlation)	Random Forest	70.030000	23.470000	12.440000	30.660000

Figure 1: Comparison of Machine Learning Models for Airbnb Price Prediction in Prague

Figure 1, compares the different ML models. Among these tested models, the Random Forest model, which uses both property and POI features, performs the best. It achieves an impressive RMSE of 68.77 and RMeSE of 24.17.

The Linear Regression model, which includes property features, POI features, spatial lag, and spatial correlation, also does well with an RMSE of 73.51 and an RMeSE of 27.68. These results highlight the importance of spatial dependencies in improving the accuracy of Airbnb price predictions.

Adding POI features significantly boosted the predictive power of both linear regression and random forest models. This shows how important proximity to points of interest is for Airbnb prices in Prague. However, the Random Forest model saw a larger performance increase, indicating its strength in capturing complex, non-linear relationships among features. Further, adding spatial dependencies to the Random Forest model did not improve the performance, suggesting that the POI features already contained the same spatial information.

In summary, these findings show that including both property-specific and geospatial features is effective for predicting Airbnb prices in Prague. It is seen that the best-performing models are either simple linear models that include complex features (property features, POI features, spatial lag of price, and spatial lag of property features) or complex models (Random Forest) that include simpler features (property features and POI features).

6 Identifying Key Drivers

For analyzing the key drivers of Airbnb prices in Prague, I used the best-performing linear regression model, which includes property features, POI features, spatial lag, and spatial correlation. This model provides insights into the relative importance of different features in predicting Airbnb prices. The table 5 demonstrates the regression coefficients for the binary property types, and table 6 for the numeric property and POI features.

Table 5: Regression Coefficients for Binary Property Types in Linear Regression Model

Feature	Coefficient	Share of RT/PT (%)
RT Shared Room	-147.794842	1.59%
PT Hotel	30.549930	3.48%
PT Hostel	-29.246329	0.96%
PT Condo	-28.169145	11.33%
PT Rental Unit	-22.586651	67.58%
PT House	-18.366693	0.70%
RT Private Room	-17.940748	14.86%
PT Apartment	8.332267	9.74%
RT Entire home/Ap	5.493882	82.44%
Spatial Lag RT Shared Room	9.233662	
Spatial Lag RT Private Room	2.204917	
Spatial Lag RT Entire home/Ap	-0.428479	
Spatial Lag PT Condo	3.672641	
Spatial Lag PT Rental Unit	3.463152	
Spatial Lag PT Hostel	2.829866	
Spatial Lag PT Hotel	-2.674588	
Spatial Lag PT House	2.093107	
Spatial Lag PT Apartment	0.717135	

Table 6: Regression Coefficients for Numeric Property and POI Features in Linear Regression Model

Feature	Coefficient
Bathrooms	18.558568
Review Scores Rating	16.335325
Accommodates	12.790124
Bedrooms	10.955701
Beds	2.148083
Spatial Lag Bathrooms	-1.872170
Spatial Lag Accommodates	-1.514050
Distance to Old Town (km)	-1.063522
Spatial Lag Bedrooms	-0.527586
Spatial Lag Beds	0.230952
Spatial Lag Review Scores Rating	0.135610
Number of Reviews	-0.113713
Spatial Lag Price	0.093473
KDE Airbnbs	0.018019
KDE Sightseeing	0.014634
KDE Shopping	-0.008843
Spatial Lag Number Reviews	0.008213
KDE Dining/Entertainment	-0.003903
KDE Transportation	-0.001660

Key influential features here include the number of bathrooms (which increases prices by about 18.56 EUR on average), review scores, property capacity (Accommodates), and number of bedrooms.

For binary property types, Shared Room listings have the most negative impact on prices, while Hotel listings have the most positive impact.

However, both of these property types have a relatively low share of the total number of listings, while Entire home/apartment listings have the highest share and a positive impact on prices.

Among the POI features, the distance to the old town has the strongest negative impact on prices, while the KDE of Airbnb's and sightseeing spots have positive impacts.

The other KDE features have almost no impact on prices in this model, which can be explained by the high positive correlation between the KDE features.

Likely, the coefficients of those features are not significant because they are already captured by the KDE of Airbnb's and Sightseeing spots. Therefore, those features could be removed from the model.

Among the Spatial Lag coefficients, the Spatial Lag of Bathrooms have the highest negative impact on prices, while the Spatial Lag of Beds have the highest positive impact on prices.

7 Conclusion

7.1 Summary of Findings

The analysis of Airbnb listings in Prague highlighted several key factors influencing property prices:

- **Location:** Listings near the city center and popular attractions command higher prices.
- **Property Size:** Larger properties with more rooms can charge higher rates as they can accommodate more guests.
- **Property Types:** Entire homes or apartments are the most popular and expensive options, with apartments being the most common type.
- **Proximity to Attractions:** Listings in areas with many dining, entertainment, and sightseeing venues are more expensive.
- **Spatial Influences:** The OLS model shows that spatial lag and correlation significantly impact prices, reflecting the influence of neighboring listings.
- **Non-Linear Relationships:** The Random Forest Regressor outperformed other models, with an RMSE of 68.77 and an RMeSE of 24.17, indicating a non-linear relationship between features and price.
- **Sensitivity to Outliers:** Despite removing outliers, the model's high RMSE compared to the RMeSE suggests it still struggles to predict outlier prices accurately.

The findings underscore the importance of location, property features, and spatial influences in shaping Airbnb pricing dynamics in Prague. Incorporating these factors into predictive models allows for a deeper understanding of property values and improve the accuracy of price predictions.

7.2 Implications and Future Research

This study has limitations, such as not considering external factors like economic conditions, political events, and global pandemics, which can significantly influence property prices. Incorporating these variables into future research could improve model predictions and provide a more comprehensive view of Airbnb pricing dynamics in Prague. Additionally, future studies could explore other geospatial features such as crime rates, neighborhood safety, and seasonal variations in pricing. These factors are crucial for understanding how external conditions and local dynamics affect Airbnb property values in the city. Lastly, this analysis is limited by data availability. Collecting more data will be essential for improving the accuracy and robustness of future price predictions.

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A Appendix

A.1 Figures

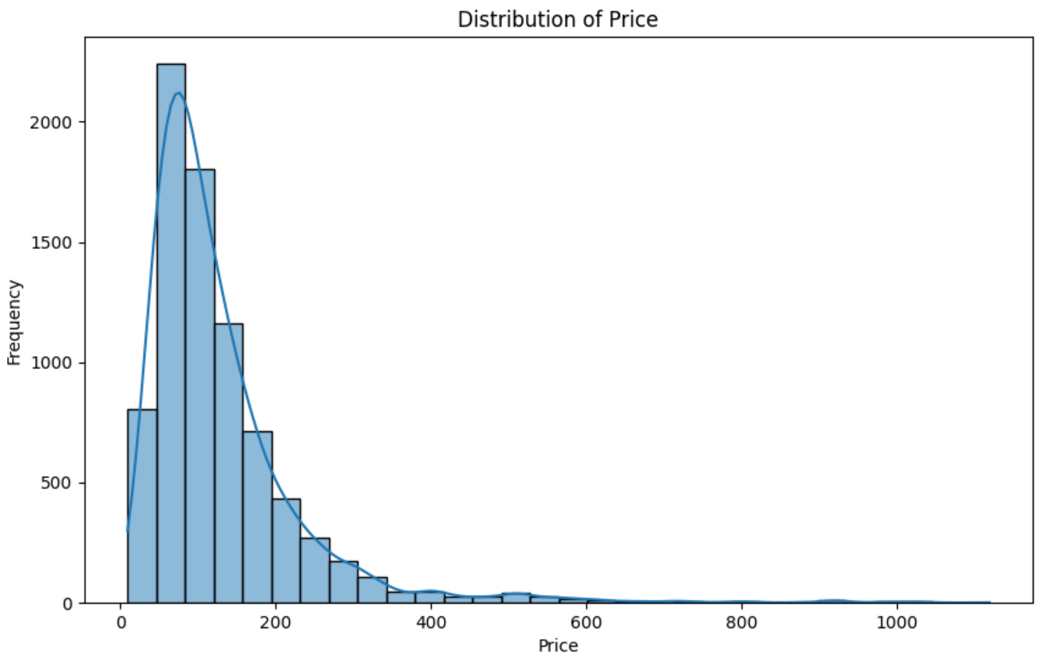

Figure 2: Histogram of Airbnb Prices(EUR) in Prague

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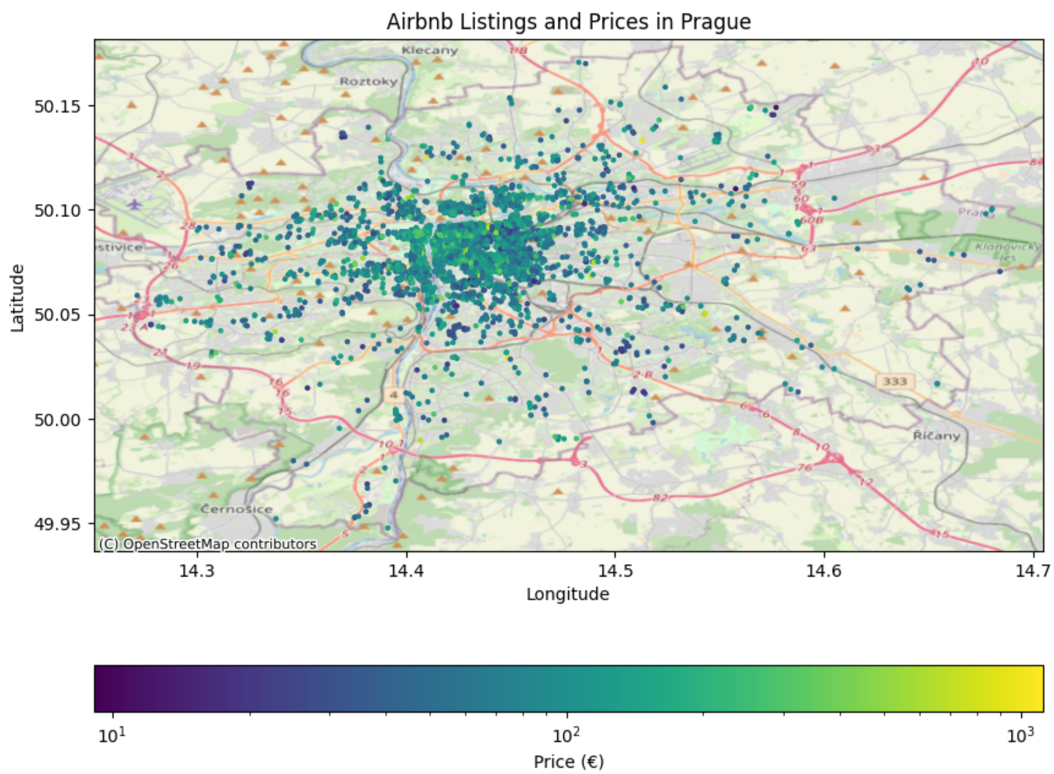

Figure 3: Spatial Distribution of Airbnb Prices in Prague

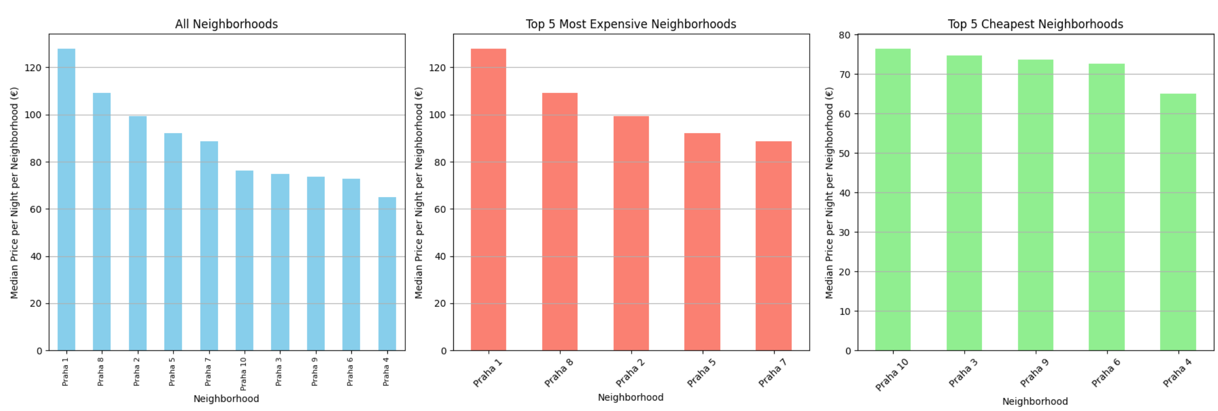

Figure 4: Median Airbnb Prices by Neighborhood in Prague

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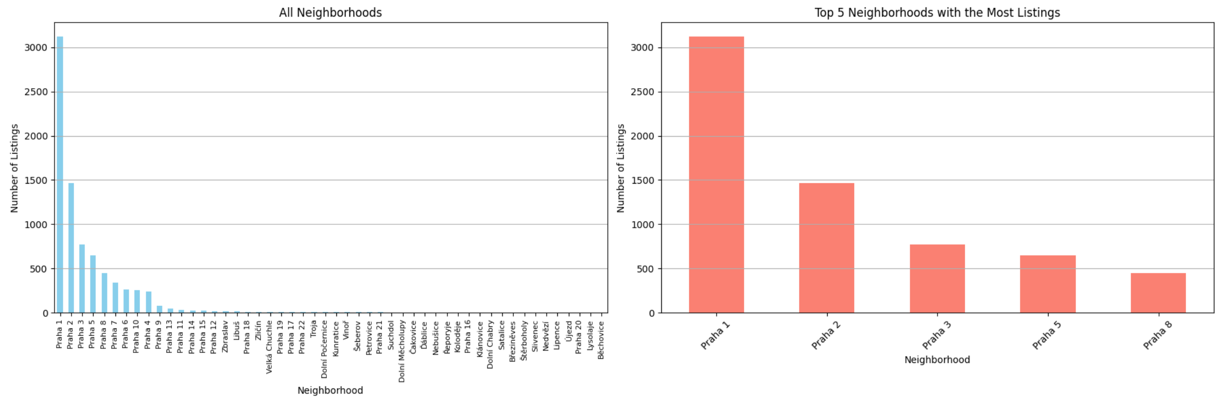

Figure 5: Number of Airbnb Listings by Neighborhood in Prague

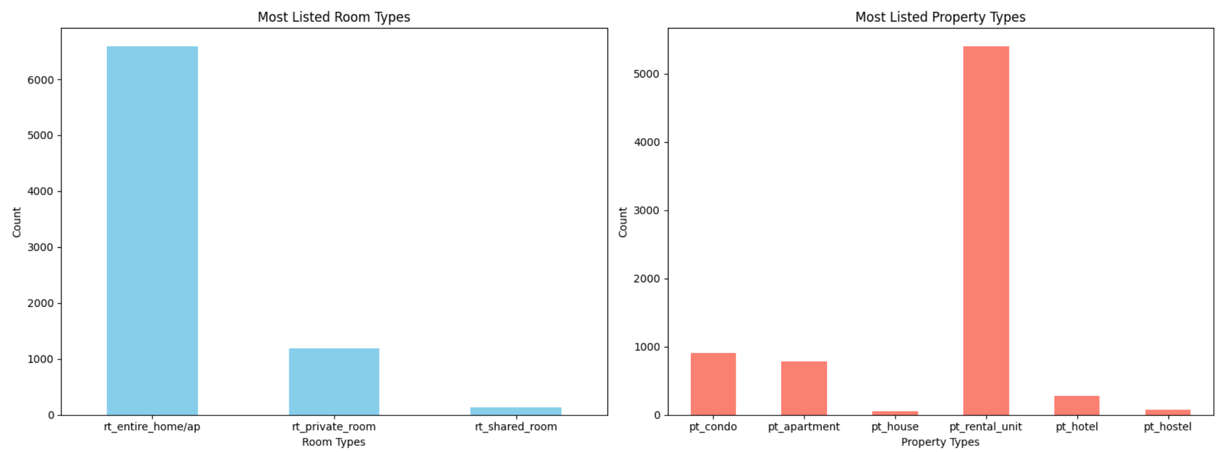

Figure 6: Most listed Room and Property Types

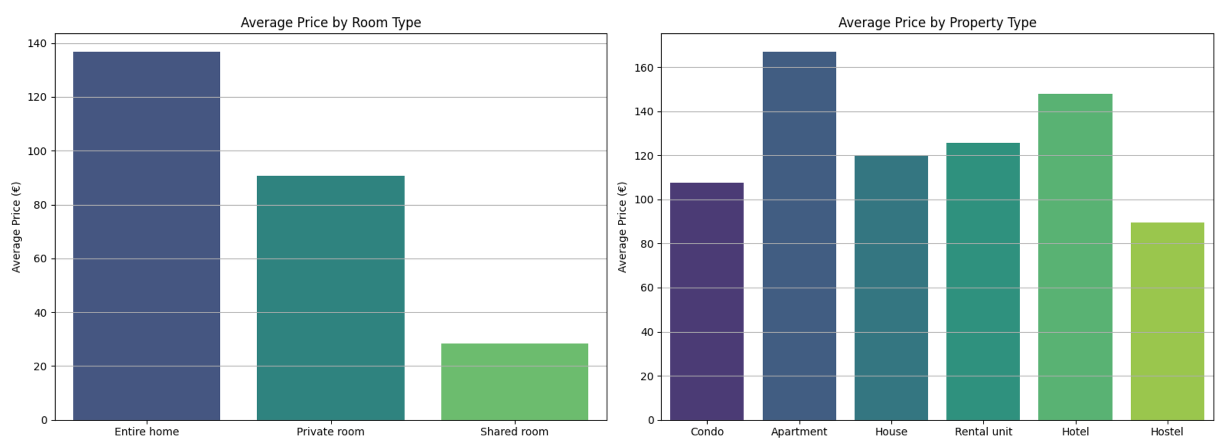

Figure 7: Average Airbnb Prices by Room and Property Type

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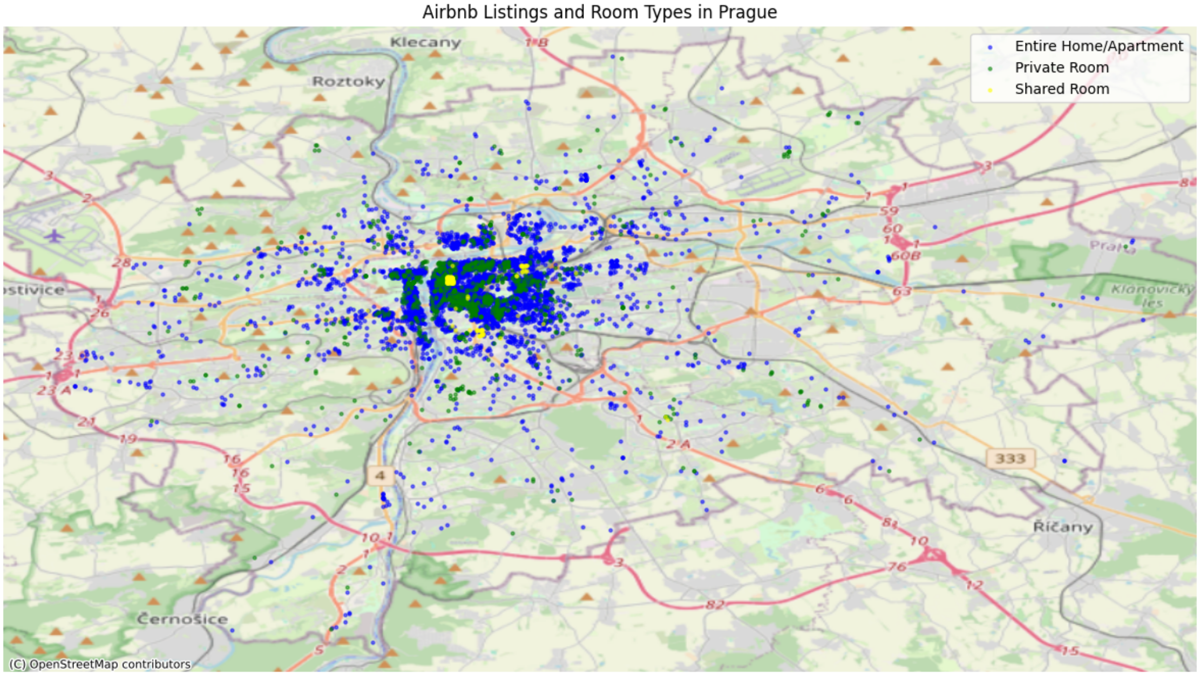
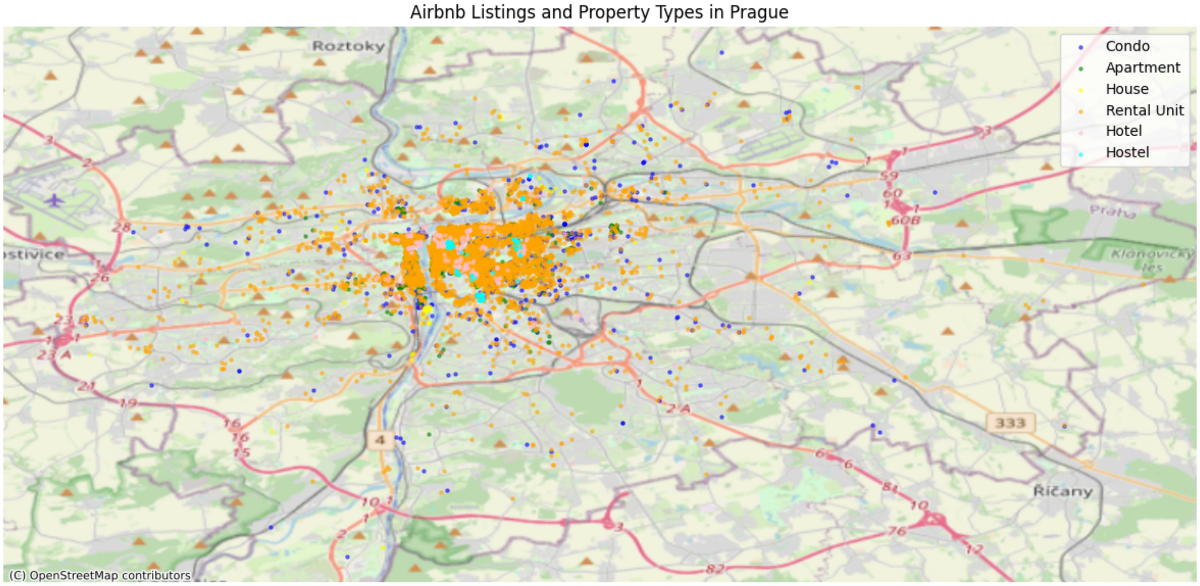
Airbnb Listings and Room Types in Prague



Airbnb Listings and Property Types in Prague



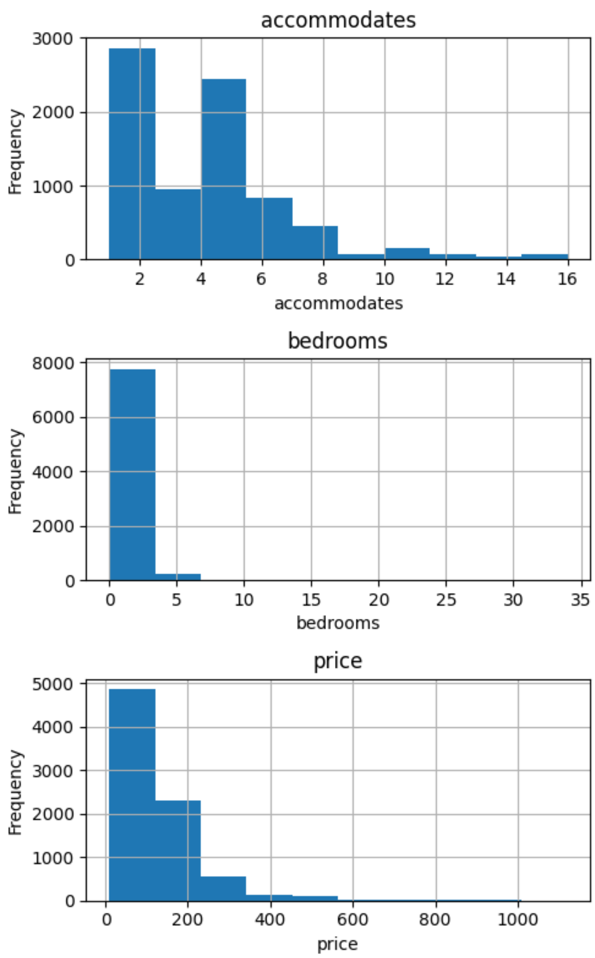


Figure 9: Frequency of Property Features

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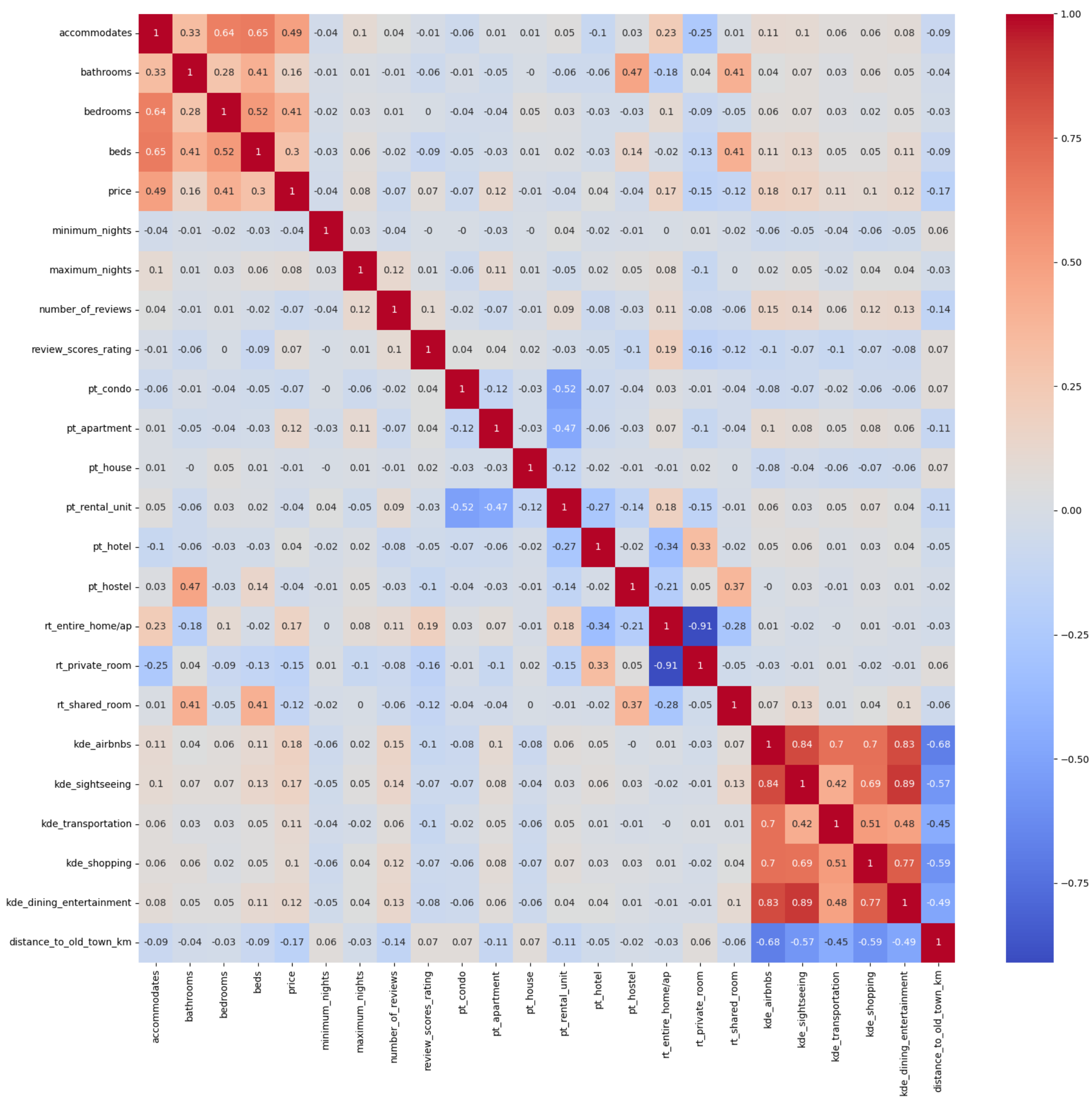

Figure 10: Correlation Matrix of Property Features, Points of Interest, and Airbnb Prices

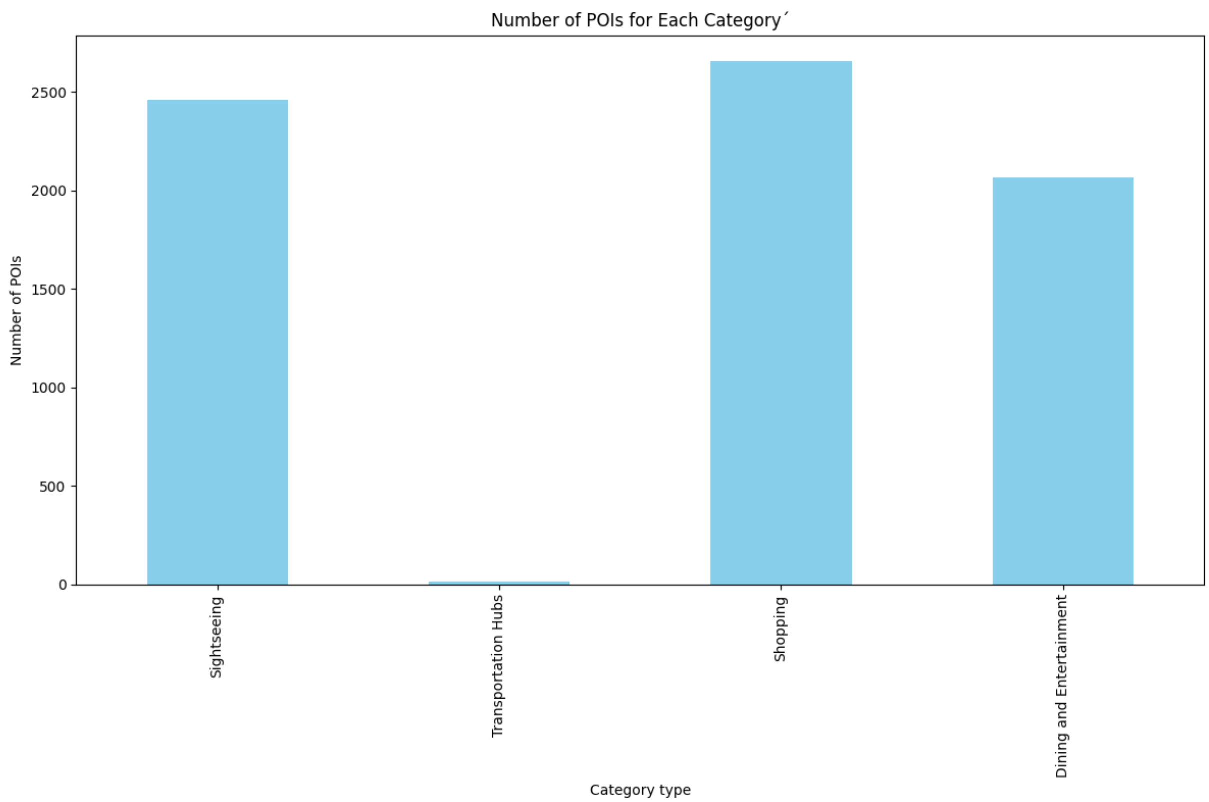

Figure 11: Frequency of Points of Interest in Prague

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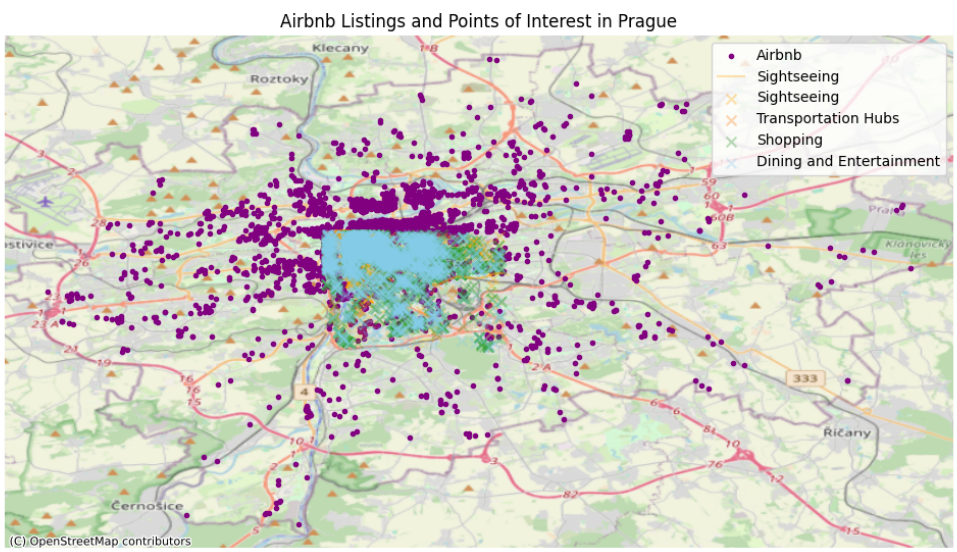

Figure 12: Spatial Distribution of Points of Interest and Airbnb Listings in Prague

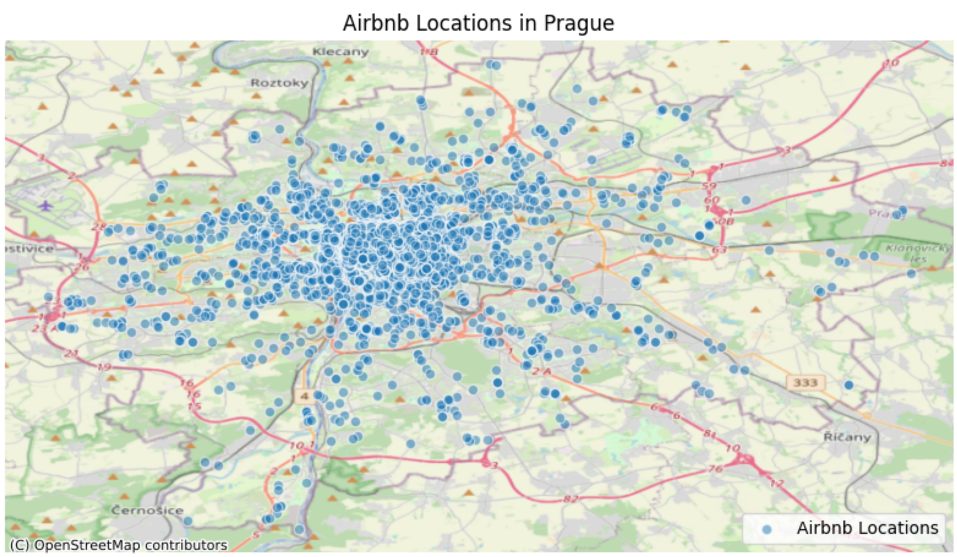

Figure 13: Distribution of Airbnb's in Prague

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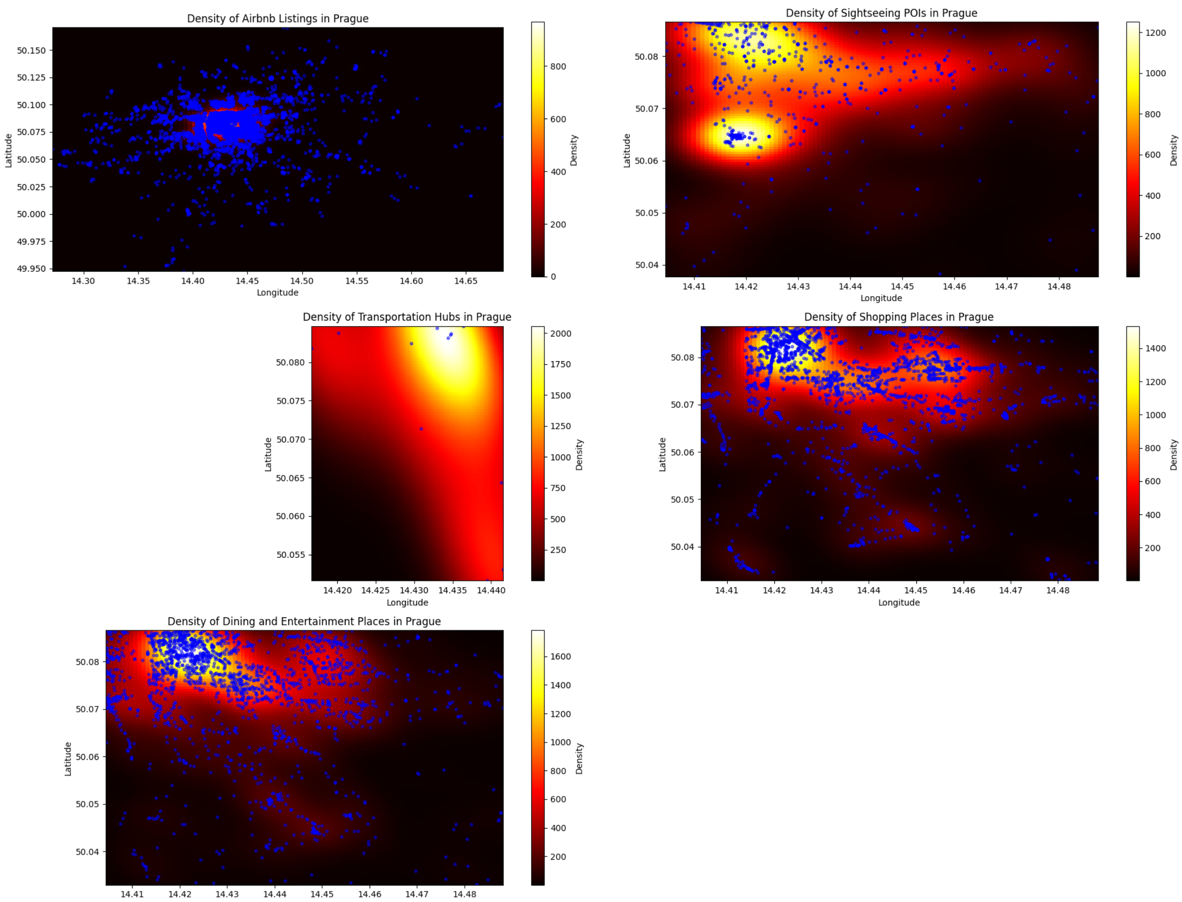

Figure 14: Kernel Density Estimation of Points of Interest in Prague

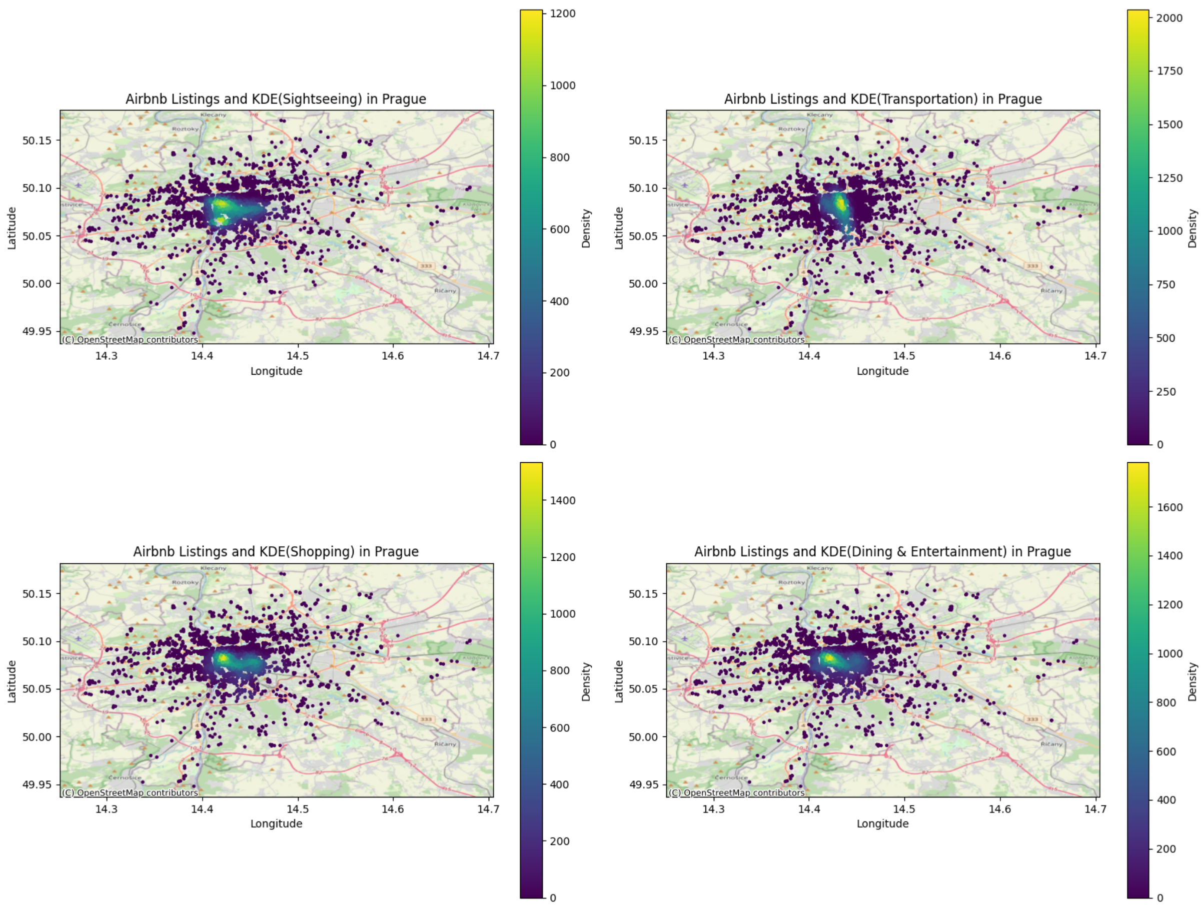

Figure 15: Kernel Density Estimation of Points of Interest and Airbnb Listings in Prague

A.2 Code

The code corresponding to this seminar thesis can be found online on Github, see: `https://github.com/luisadosch/Seminar_Airbnb`.

Hiermit versichere ich, die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie die Zitate deutlich kenntlich gemacht zu haben.
Ich erkläre weiterhin, dass die vorliegende Arbeit in gleicher oder ähnlicher Form noch nicht im Rahmen eines anderen Prüfungsverfahrens eingereicht wurde.

Würzburg, den 25. Juni 2024 Luisa Dosch

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